Final Review Lecture STAT 286

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Outline

Tablets and Test Scores

- **Goal**: estimate causal effect of providing tablets on test scores

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- Observational setting: schools decide themselves whether to adopt tablets.
- Confounding: Wealthier schools are both
  - more likely to adopt tablets, and
  - have higher test scores (tutoring, staffing, etc.).
  - Confounding in treatment **and** outcome (one would be ok).
- We suspect that

$$\mathbb{E}[Y_i \mid T_i = 1] - \mathbb{E}[Y_i \mid T_i = 0] \neq \mathbb{E}[Y_i(1) - Y_i(0)].$$

Assumption I: Super-Population

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- In the tablet example:
  - X_i: wealth characteristics of school i,
  - T_i: indicator that school i adopts tablets,
  - $Y_i(t)$: test score that would be observed for school i if $T_i = t$.

Assumption II: SUTVA

Assumption 1 (SUTVA).

- *No interference:*

$$Y_i(T_1, \ldots, T_n) = Y_i(T_i).$$

Tablet adoption of other schools don't affect school i. Student's with tablets don't move to schools without.

Assumption II: SUTVA

Assumption 1 (SUTVA).

- *No interference:*

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  Tablet adoption of other schools don't affect school i. Student's with tablets don't move to schools without.

- *Consistency:*

$$Y_i = Y_i(T_i).$$

  All tablets are treated the same. No adopted tablet is significantly different (more features etc.)

Strong Ignorability / Unconfoundedness

Assumption 2 (Strong Ignorability / Unconfoundedness).

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- Intuition: after conditioning on $\boldsymbol{X}_i$, there are **no unobserved confounders** that jointly affect T_i and Y_i.

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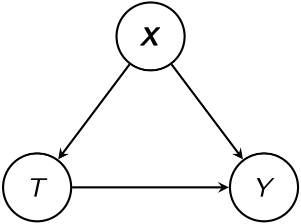
Tablet example.

- Suppose $X_i = \text{Wealth}$
- Within each wealth level, schools with and without tablets differ only by treatment, not by other unobserved factors.

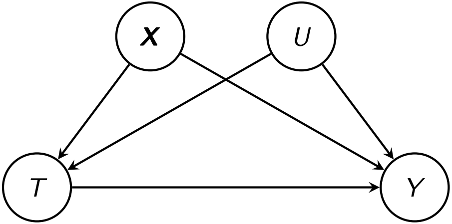
Ignorability DAG

Ignorability holds



No unmeasured confounding given $\boldsymbol{X}$.

Ignorability violated

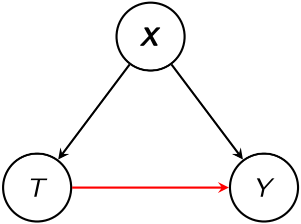


Unmeasured confounding given $\boldsymbol{X}$.

- If U exists and affects both T and Y, then $\{Y(1), Y(0)\} \not\perp\!\!\!\perp T \mid \boldsymbol{X}$; estimates that omit U are biased.
- Tablet example: U parental involvement

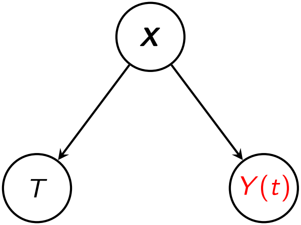
Observational / Causal DAG

Observational



T affects Y.

Causal



T affects $Y(t)$ only through $\boldsymbol{X}$.

- By setting Y to level $T = t$ ($Y \to Y(t)$), we remove the arrow from T to Y.
- All effects of T to Y go through $\boldsymbol{X}$. Conditioning *blocks* that path.
- Only condition on confounder, never on collider
- Rule of thumb: Arrow must go into Y.

Strong Overlap

Assumption 3 (Strong Overlap).

$$\exists\, \eta \in \left(0, \tfrac{1}{2}\right) \text{ s.t. } \eta < e(\boldsymbol{x}) = \Pr(T = 1 \mid \boldsymbol{X} = \boldsymbol{x}) \leq 1 - \eta \quad \text{for all } \boldsymbol{x}.$$

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Tablet example.

- With $X =$ Wealth, strong overlap requires that at every wealth level there are both tablet and non-tablet schools.

Ignorability vs. Overlap: A Tension

In observational studies, we rely on both:

- **Ignorability:** $\{Y(1), Y(0)\} \perp\!\!\!\perp T \mid \boldsymbol{X}$.
- **Overlap:** $\eta < \Pr(T = 1 \mid \boldsymbol{X}) \leq 1 - \eta$.

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- **Overlap:** $\eta < \Pr(T = 1 \mid \boldsymbol{X}) \leq 1 - \eta$.

- Adding more covariates to $\boldsymbol{X}$
  - makes ignorability more plausible (less risk of omitted confounders),
  - but makes overlap less plausible: treatment becomes more predictable and $e(\boldsymbol{X})$ gets closer to 0 or 1.

Identification: ATE

Under these assumptions we have

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where $\mu_t(\boldsymbol{x}_i) = \mathbb{E}[Y_i \mid T_i = t, \boldsymbol{X}_i = \boldsymbol{x}_i]$. We then estimate

$$\hat{\tau}^{\text{reg}}_{\text{ATE}} = \frac{1}{n}\sum_{i=1}^{n} \hat{\mu}_1(\boldsymbol{X}_i) - \hat{\mu}_0(\boldsymbol{X}_i),$$

with a *consistent* $\hat{\mu}$.

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Similarly, the ATT

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&= \mathbb{E}\big[Y_i \mid T_i = 1\big] - \mathbb{E}\big[\mu_0(\boldsymbol{X}_i) \mid T_i = 1\big].
\end{aligned}
$$

This identification strategy suggests estimating $\hat{\mu}_t$ and then forming the regression estimators

$$\hat{\tau}^{\mathrm{reg}}_{\mathrm{ATT}} = \frac{1}{n_1} \sum_{i=1}^{n} T_i(Y_i - \hat{\mu}_0(\boldsymbol{X}_i)).$$

Note that under an experimental design $\mathrm{ATE} = \mathrm{ATT}$.

Outline

Partial Identification: Big Picture

- Identification relies on strong Assumptions.
- Often, researchers doubt unconfoundedness
- Can we still say something about ATE if unconfoundedness fails?

Setup: Binary ATE Without Randomization

- Binary treatment $T_i \in \{0, 1\}$ and binary outcome $Y_i \in \{0, 1\}$.
- Potential outcomes $Y_i(0), Y_i(1)$ and the ATE

$$\tau = \mathbb{E}\big[Y_i(1) - Y_i(0)\big].$$

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- Assume:
  - i.i.d. sampling (super-population),
  - SUTVA,
  - Overlap: $0 < \Pr(T_i = 1) < 1$,
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- Assume:
  - i.i.d. sampling (super-population),
  - SUTVA,
  - Overlap: $0 < \Pr(T_i = 1) < 1$,
  - No randomization / unconfoundedness.
- Question: what can we still say about τ from observational (Y_i, T_i) alone?

Binary ATE: Decomposing the Problem

$$\mathbb{E}[Y_i(1)] = \Pr(Y_i(1) = 1 \mid T_i = 1)\Pr(T_i = 1) + \Pr(Y_i(1) = 1 \mid T_i = 0)\Pr(T_i = 0),$$
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From (Y_i, T_i) using SUTVA and Overlap we can estimate

- $\Pr(T_i = t)$
- $\Pr(Y_i(1) = 1 \mid T_i = 1) = \Pr(Y_i = 1 \mid T_i = 1)$
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However, we don't know the "cross-world" terms

$$\Pr(Y_i(1) = 1 \mid T_i = 0), \quad \Pr(Y_i(0) = 1 \mid T_i = 1)$$

are not identified without randomization.

Binary ATE: Worst-Case Bounds

- **Idea**: Bound the probabilities

$$0 \;\leq\; \Pr\big(Y_i(t) = 1 \mid T_i = 1 - t\big) \;\leq\; 1, \quad t \in \{0,1\}.$$

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- Plugging these into the decompositions yields bounds for each mean:

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&\Pr(Y_i = 1 \mid T_i = 1)\Pr(T_i = 1) \le \mathbb{E}[Y_i(1)] \le \Pr(Y_i = 1 \mid T_i = 1)\Pr(T_i = 1) + \Pr(T_i = 0),\\
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$$\Pr(Y_i = 1 \mid T_i = 0)\Pr(T_i = 0) \le \mathbb{E}[Y_i(0)] \le \Pr(T_i = 1) + \Pr(Y_i = 1 \mid T_i = 0)\Pr(T_i = 0).$$

- Combining worst-case upper and lower bounds yields

$$\tau_{\text{min}} \;\le\; \tau \;\le\; \tau_{\text{max}},$$

where

$$\tau_{\text{max}} = \Pr(Y_i = 1 \mid T_i = 1)\Pr(T_i = 1) + \Pr(Y_i = 0 \mid T_i = 0)\Pr(T_i = 0),$$
$$\tau_{\text{min}} = -\Pr(Y_i = 0 \mid T_i = 1)\Pr(T_i = 1) - \Pr(Y_i = 1 \mid T_i = 0)\Pr(T_i = 0).$$

IV Setting: Setup and DAG

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- Binary encouragement $Z_i \in \{0,1\}$, binary treatment $T_i \in \{0,1\}$, binary outcome $Y_i \in \{0,1\}$.
- IV assumptions:
  - Randomization of Z: $Z_i \perp\!\!\!\perp \big(T_i(z), Y_i(z,t)\big)$ for all (z,t).
  - Exclusion restriction: $Y_i(z,t) = Y_i(z',t)$ for all z, z', t,
  - No monotonicity or relevance assumptions.

$$\begin{tikzcd}
 & & U \arrow[dl] \arrow[dr] & \\
Z \arrow[r] & T \arrow[rr] & & Y
\end{tikzcd}$$

IV Setting: Setup and DAG

- Binary encouragement $Z_i \in \{0,1\}$, binary treatment $T_i \in \{0,1\}$, binary outcome $Y_i \in \{0,1\}$.
- IV assumptions:
  - Randomization of Z: $Z_i \perp\!\!\!\perp \big(T_i(z), Y_i(z,t)\big)$ for all (z,t).
  - Exclusion restriction: $Y_i(z,t) = Y_i(z',t)$ for all z, z', t,
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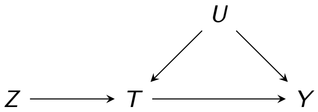

- Goal: bounds on the ATE (not LATE!)

$$\tau = \mathbb{E}\big[Y_i(1) - Y_i(0)\big]$$

when T is confounded by U but Z is a valid instrument.

IV Setting: Latent Types and the Estimand

- Define the *latent type* for unit i:

$$U_i = \big(T_i(0), T_i(1), Y_i(0), Y_i(1)\big) \in \{0, 1\}^4,$$

so there are 16 possible types u.

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$$y_1(u) = \begin{cases} 1, & \text{if } u = (t_0, t_1, y_0, y_1) = (t_0, t_1, y_0, 1), \\ 0, & \text{otherwise} \end{cases}$$

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- So τ is a *linear function* of the unknown probabilities π_u.

IV Setting: Observed Data and Constraints

We observe the joint distribution

$$\Pr(Y_i = y, T_i = t, Z_i = z) \quad \text{for } y, t, z \in \{0, 1\}.$$

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IV Setting: Sharp Bounds via a Linear Program

This leads to a pair of linear programs in the unknowns $\Pr(U_i = u)$:

$$\text{maximize / minimize } \tau = \sum_u (\mathbb{1}\{y_1(u) = 1\} - \mathbb{1}\{y_0(u) = 1\}) \Pr(U_i = u)$$

subject to

$$\Pr(Y_i = y, T_i = t, Z_i = z) = \sum_u \mathbb{1}\{y_t(u) = y, t_z(u) = t\} \Pr(Z_i = z) \Pr(U_i = u)$$

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How to solve this:

1. Derive bounds on τ within the constraints
2. Tip: make sure that the probability bounds are satisfied.
3. Show that these are *sharp*: there exists a probability combination of $\{\Pr(U_i = u)\}_u$ that achieves these bounds.

Outline

Panel Data: Missing Outcome Problem

- We now observe the *same* units repeatedly over time $\{Y_{it}\}_{i=1,\ldots,N,t=1,\ldots,T}$.
- Interested in ATT
- All units start untreated; at some later time, a subset becomes treated

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- Interested in ATT
- All units start untreated; at some later time, a subset becomes treated
- For a newly treated unit i at time $t+1$, the causal effect is

$$\tau_{it+1} = Y_{it+1}(1) - Y_{it+1}(0) = Y_{it+1} - Y_{it+1}(0)$$

- Need to impute: $Y_{it+1}(0)$

Naive Time Comparison: Before–After

- **Before–after idea:** compare the same unit over time:

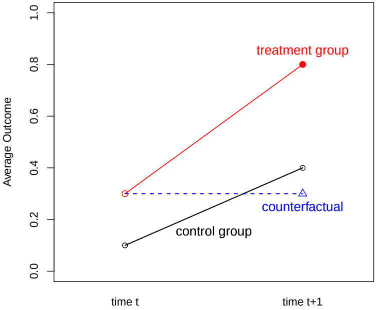
$$\hat{\tau}_{it+1} = Y_{it+1} - Y_{it}.$$

- This assumes that

$$Y_{it+1}(0) = Y_{it}(0),$$

i.e. the untreated outcome would not change over time.

- Unrealistic: ignores common trends, macro shocks, aging, etc.



Before–after comparison (taken from Kosuke Imai's lecture slides).

Naive Cross-Sectional Comparison

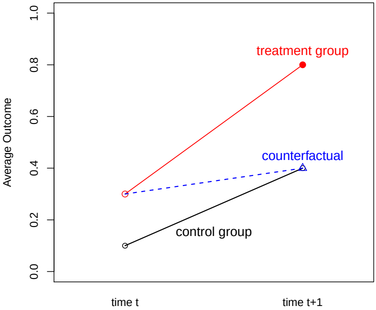
- **Cross-sectional idea** use units that remain in the control group at time $t + 1$.
- Compare treated unit i to a control unit j at the same time:

$$\hat{\tau}_{it+1} = Y_{it+1} - Y_{jt+1}.$$

- For treated unit i and control unit j, this assumes

$$Y_{it+1}(0) = Y_{jt+1}(0),$$

  i.e. they would have had the same untreated outcome at $t + 1$.
- Unrealistic: treatment is not randomized, systematic differences between treated and control units (selection into treatment)



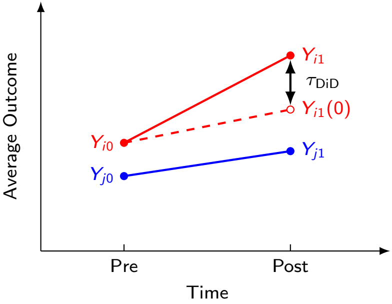
Cross-sectional comparison (taken from Kosuke Imai's lecture slides).

Difference-in-Differences: Combining Time and Units

- **Idea:** combine the before–after and cross-sectional comparisons.
- **Parallel trends assumption:**

$$\mathbb{E}\left[Y_{i1}(0) - Y_{i0}(0)\right] = \mathbb{E}\left[Y_{j1}(0) - Y_{j0}(0)\right],$$

  i.e. in the absence of treatment, treated and control units would follow the same average trend.
- $\tau_{\text{DiD}} = \left(Y_{i1} - Y_{i0}\right) - \left(Y_{j1} - Y_{j0}\right)$.
- Imputes:
  $Y_{it+1}(0) = Y_{i0}(0) + (Y_{j1}(0) - Y_{j0}(0))$,
  i.e, before treatment + trend of control



Difference-in-differences in the 2×2 case.

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- Causal estimand: the individual effect on unit N at time T,

$$\tau_N = Y_{NT}(1) - Y_{NT}(0) = Y_{NT} - Y_{NT}(0).$$

- Problem: $Y_{NT}(0)$ is unobserved $\Rightarrow$ need to impute it.

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- The treated unit's untreated path is a time-invariant weighted average of control paths.

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  - Unconstrained regression can predict far outside the range of the data.

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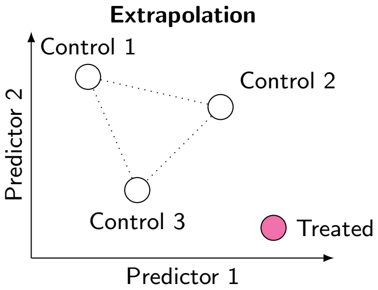
- **Extrapolation:**
  - The treated unit lies *outside* the convex hull of controls.
  - Unconstrained regression can predict far outside the range of the data.
- **Interpolation:**
  - Project the treated unit into the convex hull of controls and predict *within* a plausible region.
  - This guards against wild predictions when we only care about a single treated unit.



Extrapolation (left) predicts using controls outside their convex hull; interpolation (right) projects into the convex hull of controls.

Interpolation vs. Extrapolation

- Example:
  - All controls have unemployment between 5% and 10%.
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- Example:
  - All controls have unemployment between 5% and 10%.
  - Unconstrained regression could predict 20%.
  - Synthetic control restricts predictions to $[5\%, 10\%]$.
- As a prediction method for a single trajectory, avoiding extrapolation is crucial.

Fitting the Synthetic Control

- Choose $\hat{w} = (\hat{w}_1, \ldots, \hat{w}_{N-1})$ to match the pre-treatment trajectory of unit N:

$$\hat{w} = \arg\min_{w} \sum_{t=1}^{T-1} \left(Y_{Nt} - \sum_{i=1}^{N-1} w_i Y_{it} \right)^2$$

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- Works best with long, stable pre-treatment trajectories and a large donor pool.

Question

Important topics not discussed:

- RDD
- Double Robust Estimation
- Mediation

Any Questions?

Good luck!

You've got this! Go show off those causal superpowers.